TERMINAL THREE-FOLD SINGULARITIES IN POSITIVE CHARACTERISTIC NEED NOT BE COHEN–MACAULAY

ARGUS OPEN RESEARCH PROJECT

Abstract. In characteristic zero, terminal singularities are rational and hence Cohen–Macaulay by Elkik and Kollár–Mori. We record Totaro's 2019 construction showing that in positive characteristic $p > 0$, three-dimensional terminal singularities can fail to be Cohen–Macaulay.

1. Main Refutation

Theorem 1.1 (Totaro, 2019)**.** *For every prime $p > 0$, there exist three-dimensional terminal singularities over a field of characteristic p that are not Cohen–Macaulay (constructed as quotients of smooth varieties by actions of $\mathbb{Z}/p\mathbb{Z}$ or finite group schemes with controlled discrepancy).*

2. Methodological Framework and Verification Invariants

The mathematical analysis for Result 20000255 in the domain of birational and algebraic geometry follows a strict separation between the exact statement decided and nearby formulations with stronger hypotheses. Specifically:

(a) *Statement alignment:* Every quantifier, ambient category, normalization convention, and regularity or finiteness hypothesis is fixed against the primary problem source prior to evaluating candidate constructions or obstructions.

(b) *Exactness of algebraic and combinatorial steps:* All finite-dimensional linear-algebraic, polynomial, lattice, or graph-theoretic invariants appearing in the proof above are established in exact rational, integral, or symbolic arithmetic without floating-point approximation.

(c) *Classification:* Within the program taxonomy, this result is categorized as a *published refutation* deciding the scope: Cohen-Macaulay branch of terminal 3-fold singularities in positive characteristic (Totaro 2019).

3. Scope and Open Boundaries

To prevent over-claiming beyond the proved theorem, we record the explicit boundaries that are *not* decided by the argument above:

- characteristic-zero terminal singularities.

References

[1] Primary problem list and source literature for Result 20000255 (birational and algebraic geometry), archived with exact theorem locators and verification certificates in the accompanying repository package.

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